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Third SEMESTER–B.TECH- Course Work
MID-SEMESTER EXAMINATION, September 2026

Course Code-EEMTC04/ICMTC04
Course Title-Mathematics-III
Time: 1:30 hours Max Marks-20

Note: Attempt all questions. Missing data/information (if any), may be suitably assumed & mentioned in the answer.

Q. No.	Question	Marks	CO
1 (a)	Show that the function $u(x,y) = 4xy - 3x + 2$ is harmonic.	2	CO1
(b)	Test the analyticity of the function $w = sinz$.	2	CO1
2 (a)	Evaluate the following integral $\int_C \frac{1}{z}\cos z dz$, where C is the ellipse $9x^2 + 4y^2 = 1$.	2	CO1
(b)	Determine the poles and the residue at each pole of the function $f(z) = \frac{z^2}{(z-1)^2(z+2)}$.	2	CO1
3 (a)	Show that the real and imaginary part of the function $w = \log z$ satisfy the Cauchy-Reimann equations when z is not zero.	2	CO1
(b)	Determine the constant a so that the vector $\vec{F} = (x+3y)\hat{i} + (y-2z)\hat{j} + (x+az)\hat{k}$ is solenoidal.	2	CO2
4 (a)	Evaluate $\int_C [(2x^2y + y + z^2)\hat{i} + 2(1+yz^3)\hat{j} + (2z + 3y^2z^2)\hat{k}].d\vec{r}$, along the curve $y^2 + z^2 = a^2$, $x = 0$.	2	CO2
(b)	Determine whether $\vec{F} = (y^2\cos x + z^3)\hat{i}$ $+(2y\sin x - 4)\hat{j} + (3xz^2 + 2)\hat{k}$, is a conservative vector field.	2	CO2
5 (a)	Evaluate by Green's theorem, $\int_C e^{-x}(\sin y\, dx + \cos y\, dy)$ where C is the rectangle with vertices (0,0), $(\pi, \frac{\pi}{2})$, $(0, \frac{\pi}{2})$ and $(\pi, 0)$.	2	CO2
(b)	Evaluate $\iint_S \vec{F}.dS$, where $\vec{F} = x\hat{i} - (z^2 - zx)\hat{j} - xy\hat{k}$ and S is the triangular surface with vertices (2,0,0), (0,2,0) and (0,0,4).	2	CO2